

PRODUCT BRIEF

TradePilot

*The platform that makes Indian traders
profitable*



Version: v0.1.0 (MVP)

Status: Planning

Created: April 2, 2026

Target Launch: 8 weeks from sprint start

TradePilot -- Product Brief

1. What We're Building

A mobile + web trading platform for beginner Indian traders that shows **AI-powered profit probability before every trade**. Integrated with Zerodha via Kite Connect API. Built with Flutter (frontend) + Rust (backend) + QuestDB (market data).

1.1 MVP Scope -- What's IN

FEATURE	PRIORITY	SPRINT
AI Trade Scorer (profit probability per trade)	P0	3
Zerodha Kite Connect integration (auth + orders)	P0	1
Real-time market data (WebSocket streaming)	P0	1
Stock search + detail screen with charting	P0	2
Order placement (market, limit, SL, SL-M)	P0	2
Portfolio dashboard (positions, holdings, P&L)	P0	4
Watchlist management	P1	2
Risk guardrails (max loss alert, position sizing)	P1	3
Basic educational tooltips for beginners	P1	4

1.2 What's OUT (v0.2+)

- Paper trading / simulator
- Backtesting engine
- Options strategy builder with Greeks
- Full educational content library
- Multi-broker support (Angel One)

- Social/community features
 - Tax reporting
 - Advanced charting (custom indicators)
 - Push notifications
 - Fractional investing
-

2. User Flow

2.1 First-Time User Journey

ONBOARDING (2 min)

1. Download app / open web
2. Sign up (phone + OTP)
3. "Connect your Zerodha account"
 - > Redirect to Zerodha login (OAuth flow)
 - > Grant permissions -> return to TradePilot
4. Welcome screen: "Your trading co-pilot is ready"

DISCOVER (browsing)

5. Dashboard loads: Market overview + trending stocks
6. Browse watchlist or search for a stock
7. Tap stock -> Stock Detail screen

The Money Moment -- AI Trade Scorer

TRADE SCORER

RELIANCE @ Rs 2,845

AI Profit Score: **72%** [=====>]

Suggested Action: **BUY** (short-term bullish)

Risk/Reward: 1:2.4

Recommended Size: 5 shares (Rs 14,225)

Max Risk: Rs 1,500 (your daily limit)

Why this score:

- FII net buyers +1,200 Cr today
- RSI oversold (28.3), bounce likely
- Sector momentum: Energy +1.8% today
- Similar setups: 72% profitable (90-day)

BUY NOW

ADD TO WATCHLIST

LEARN MORE

TRADE (execution)

9. Tap "BUY NOW"
10. Order form pre-filled with recommended qty + SL
11. Confirm -> order placed via Zerodha API
12. See confirmation + position added to portfolio

TRACK (post-trade)

13. Portfolio shows live P&L
14. AI alerts if score changes significantly
15. "Consider exiting" nudge when target hit

2.2 Daily Returning User

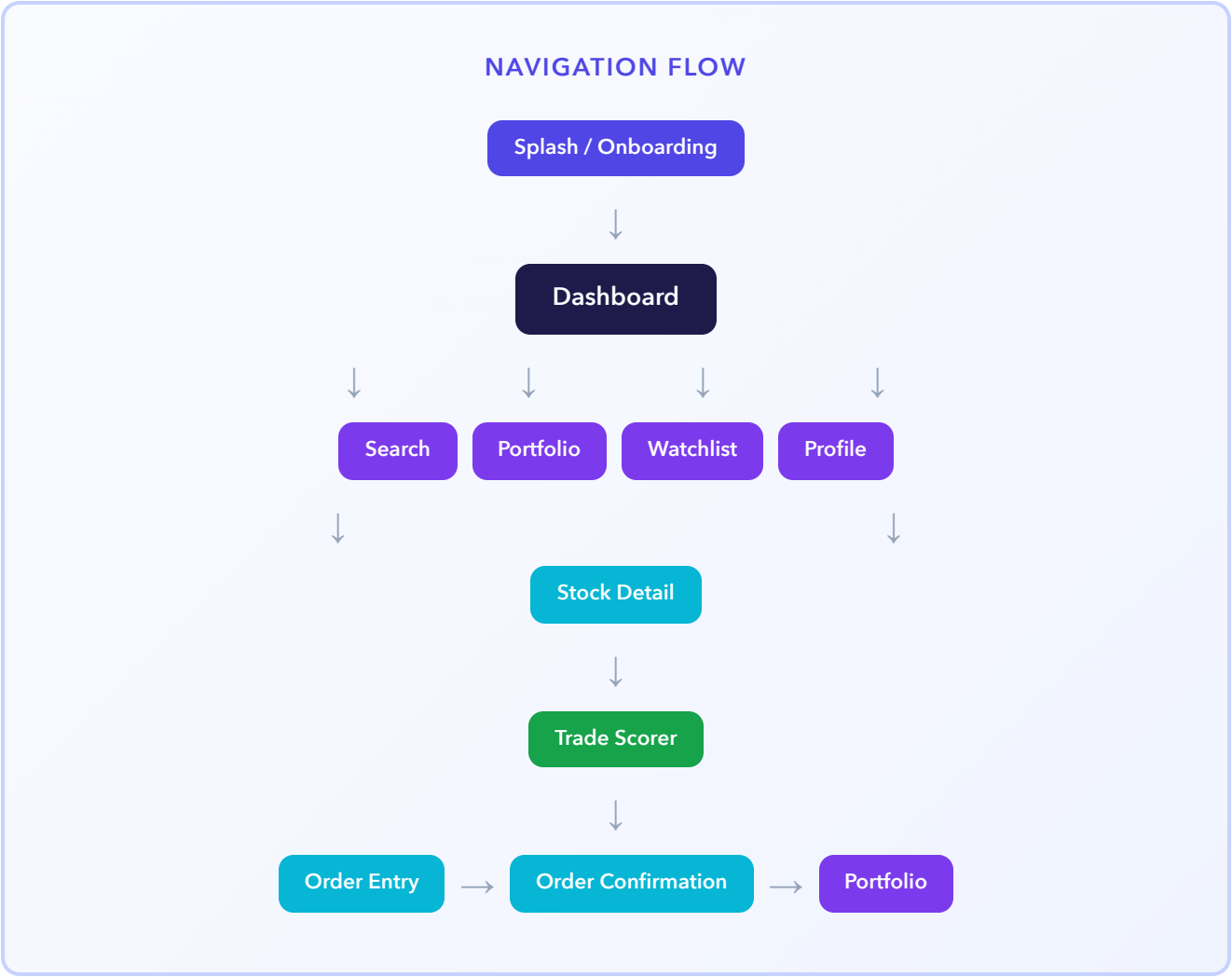
1. Open app -> Dashboard
 2. See portfolio P&L (green/red)
 3. AI "Today's Opportunities" card (top 5 scored trades)
 4. Tap any -> Trade Scorer -> decide -> trade or skip
 5. Check watchlist for tracked stocks
-

3. Screen List (11 screens)

3.1 Core Screens

#	SCREEN	PURPOSE	KEY ELEMENTS
1	Splash / Onboarding	First launch, sign up	Phone OTP, Zerodha connect
2	Dashboard	Home screen, market overview	Market indices, portfolio summary, AI picks, watchlist
3	Search	Find stocks/indices	Search bar, trending, sectors, recent
4	Stock Detail	Single stock deep dive	Chart (TradingView), fundamentals, news, AI score card
5	Trade Scorer	AI profit probability	Score gauge, reasons, recommended action/size/SL
6	Order Entry	Place a trade	Pre-filled from scorer, order type, qty, price, SL
7	Order Confirmation	Post-trade confirmation	Order ID, status, position link
8	Portfolio	Positions + holdings	Live P&L, positions, holdings, day's trades
9	Watchlist	Track stocks	Saved stocks with mini AI scores
10	Profile / Settings	Account, broker connection	Zerodha status, risk limits, preferences
11	Learn	Educational tooltips	Contextual learning cards (what is SL, P&L, etc.)

3.2 Screen Flow Diagram



4. Data Model

4.1 PostgreSQL (User data, application state)

```
-- Users
CREATE TABLE users (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    phone VARCHAR(15) UNIQUE NOT NULL,
    name VARCHAR(100),
    email VARCHAR(255),
    created_at TIMESTAMPTZ DEFAULT NOW(),
    updated_at TIMESTAMPTZ DEFAULT NOW()
);

-- Broker connections
CREATE TABLE broker_connections (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID REFERENCES users(id),
    broker VARCHAR(20) NOT NULL DEFAULT 'zerodha',
    access_token TEXT,
    refresh_token TEXT,
    token_expires_at TIMESTAMPTZ,
    zerodha_user_id VARCHAR(20),
    is_active BOOLEAN DEFAULT true,
    connected_at TIMESTAMPTZ DEFAULT NOW()
);

-- Trade scores (AI predictions)
CREATE TABLE trade_scores (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    instrument_token BIGINT NOT NULL,
    trading_symbol VARCHAR(50) NOT NULL,
    score DECIMAL(5,2) NOT NULL,      -- 0-100 profit probability
    direction VARCHAR(10) NOT NULL,  -- 'buy' or 'sell'
    risk_reward_ratio DECIMAL(5,2),
    recommended_qty INTEGER,
    recommended_sl DECIMAL(12,2),
    reasons JSONB,
    model_version VARCHAR(20),
    confidence DECIMAL(5,2),
    scored_at TIMESTAMPTZ DEFAULT NOW()
);
```

```
-- Watchlists, Risk Settings, Order History
-- (Additional tables for watchlist_items,
-- user_risk_settings, order_history)
```

4.2 QuestDB (Market data, time-series)

```
-- OHLC candle data (from Zerodha historical API)
CREATE TABLE ohlc (
    timestamp TIMESTAMP,
    instrument_token LONG,
    trading_symbol SYMBOL,
    open DOUBLE, high DOUBLE, low DOUBLE, close DOUBLE,
    volume LONG
) TIMESTAMP(timestamp) PARTITION BY DAY;

-- Live quotes (WebSocket), Indicators (computed)
-- RSI, MACD, SMA, EMA, ATR, VWAP partitioned by DAY
```

5. AI Trade Scorer -- How It Works

5.1 Input Features (per stock)

CATEGORY	FEATURES	SOURCE
Price action	RSI, MACD, SMA crossovers, Bollinger Bands	Computed from OHLC
Volume	Volume vs 20-day avg, OBV, volume spikes	Computed from OHLC
Trend	SMA 20/50/200 alignment, ADX, price vs VWAP	Computed from OHLC
Volatility	ATR, historical volatility, day range %	Computed from OHLC
Market context	NIFTY trend, sector performance, VIX level	NSE data
Institutional	FII/DII net buy/sell (daily)	NSE published data
Historical	Similar setups in last 90/180/365 days, win rate	QuestDB queries

5.2 Scoring Pipeline

Every 5 minutes during market hours:

1. Fetch latest OHLC (1-min candles) from QuestDB
2. Compute technical indicators (RSI, MACD, SMA, etc.)
3. Fetch market context (NIFTY, VIX, FII/DII)
4. Run ML model (XGBoost for v0.1, upgrade to LSTM later)
5. Output: probability score (0-100) + direction + confidence
6. Store in trade_scores table
7. Push to connected clients via WebSocket

On user request (stock detail page):

1. Return latest score from trade_scores
2. If score older than 5 min, recompute on-demand
3. Include top 4 reasons (from feature importance)

5.3 Model Architecture (v0.1)

Why XGBoost, not deep learning for MVP

- Explainable (feature importance = "reasons" for the score)
- Fast inference (< 10ms per stock)
- Works well with tabular data
- Trainable on limited data
- Upgrade path to LSTM/Transformer in v0.2

Training data: Historical OHLC for NIFTY 500 stocks (5 years daily, 1 year intraday). Label: "Was this trade profitable?" (1-day, 3-day, 5-day forward returns). Features: 20+ technical indicators + market context.

5.4 Risk Guardrails

GUARDRAIL	WHAT IT DOES	DEFAULT
Max daily loss	Alert when cumulative day loss hits limit	Rs 5,000
Position size cap	Max % of capital per trade	10%
Score threshold	Warn if placing trade with score < 40	40%
Volatility warning	Extra caution flag on high-VIX days	VIX > 20

6. Zerodha Integration Details

6.1 Auth Flow

1. User taps "Connect Zerodha"
2. Redirect to: `https://kite.zerodha.com/connect/login?v=3&api_key={KEY}`
3. User logs in + grants permission
4. Zerodha redirects back with `request_token`
5. Backend: `POST /session/token` with `SHA256(api_key + request_token + api_secret)`
6. Receive `access_token` -> store in `broker_connections`
7. Token valid until ~6 AM next day -> auto-refresh daily

6.2 API Usage Map

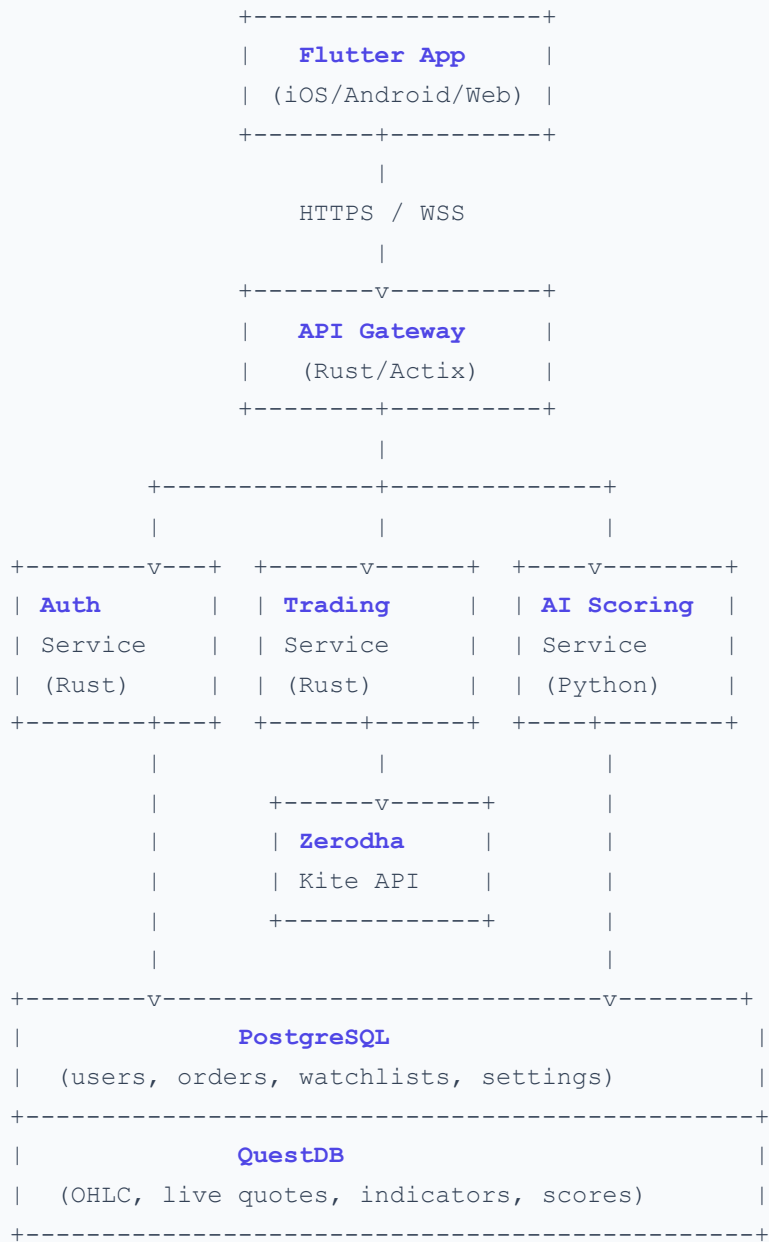
TRADEPILOT FEATURE	KITE API ENDPOINT	RATE IMPACT
Portfolio positions	<code>GET /positions</code>	1 req/refresh
Holdings	<code>GET /holdings</code>	1 req/refresh
Place order	<code>POST /orders/{variety}</code>	1 req/order
Live quotes	WebSocket (3K instruments)	Streaming
Historical OHLC	<code>GET /instruments/historical</code>	Batch nightly
Search instruments	<code>GET /instruments</code> (CSV)	1 req/day
User profile	<code>GET /user/profile</code>	1 req/session
Margins	<code>GET /user/margins</code>	1 req/trade

6.3 Constraints to Design Around

CONSTRAINT	IMPACT	OUR SOLUTION
Token expires daily ~6 AM	User must re-auth	Auto-detect expired token, prompt re-login
10 req/sec rate limit	Can't poll aggressively	WebSocket for live data, batch REST calls
3,000 orders/day	Not an issue for retail	Monitor and warn at 2,500
No bracket orders	Can't do entry + SL + target in one	Place separate SL order after main order
No tick data	Can't build tick-level models	Use 1-min candles as finest granularity
Historical data limits	60 days for 1-min candles	Nightly batch download, store in QuestDB

7. Tech Architecture

7.1 System Overview



7.2 Rust Backend Crates

CRATE	RESPONSIBILITY
<code>tradepilot-api</code>	Actix-web API gateway, routes, middleware
<code>tradepilot-auth</code>	JWT auth, Zerodha OAuth, session management
<code>tradepilot-trading</code>	Order management, position sync, Kite API client
<code>tradepilot-market</code>	WebSocket manager, quote streaming, OHLC ingestion
<code>tradepilot-scoring</code>	Score cache, serving layer, Python bridge (PyO3)
<code>tradepilot-models</code>	Shared types, DB models, DTOs
<code>tradepilot-db</code>	PostgreSQL + QuestDB connection pools

7.3 Flutter App Structure

```
lib/  
  main.dart  
  app/  
    routes.dart, theme.dart  
  features/  
    auth/          -- login, Zerodha connect  
    dashboard/     -- home screen, market overview  
    search/        -- stock search  
    stock_detail/  -- chart, fundamentals, AI score  
    trade_scorer/  -- profit probability UI  
    order/         -- order entry, confirmation  
    portfolio/     -- positions, holdings, P&L  
    watchlist/     -- manage watchlists  
    settings/      -- profile, risk limits  
    learn/         -- educational tooltips  
  core/  
    api/           -- API client (Dio)  
    websocket/     -- live data streaming  
    models/        -- data models  
    providers/     -- Riverpod state management  
    widgets/       -- shared UI components
```

8. Design Principles

PRINCIPLE	WHAT IT MEANS
Beginner-first	Every screen should be understandable by someone who's never traded
Score-led	AI Trade Score is visible everywhere -- dashboard, search results, watchlist
Safe by default	Risk guardrails ON by default, user must opt out (not opt in)
Fast	App loads in < 2 seconds, scores update in < 5 seconds
Honest	"72% probability" not "guaranteed profit". Always show confidence level
Contextual learning	"What is RSI?" tooltip appears when RSI is shown, not in a separate section

9. Success Metrics (6-month target)

METRIC	TARGET	HOW WE MEASURE
Paying users (Pro)	1,000	Subscription count
Daily active users	3,000	App opens with trade/view activity
Avg trades per user/week	5+	Order count / active users
Trade score accuracy	> 60%	Scored trades that were profitable
App rating	> 4.2	App Store / Play Store
Churn (monthly)	< 8%	Users who stop opening app
NPS	> 40	In-app survey

Key Differentiator: TradePilot is not another charting app or brokerage clone. It is the first Indian trading platform that tells you the **probability of profit before you trade**, making it accessible to the 90% of retail traders who currently lose money.